

INTERNSHIP EVALUATION REPORT

Walmart USA – Advanced Software Engineering Virtual Experience (Forage)

Student Name:	Deepanshu	Academic Term:	Odd Semester (5th Sem)
Degree Program:	B.Tech Computer Science & Engineering	Domain:	Software Engineering & Data Systems

1. EXECUTIVE SUMMARY

This report summarizes the technical tasks, architectural designs, and data pipelines completed during the **Walmart USA Advanced Software Engineering Job Simulation** on Forage. The internship focused on solving complex backend, data structure, software architecture, relational database design, and ETL (Extract, Transform, Load) challenges tailored to retail logistics and high-throughput enterprise processing systems.

2. TECHNICAL TASKS & IMPLEMENTATION DETAILS

Task 1: Advanced Data Structures (\$2^x\$ Max Heap)

Objective: Design and implement a non-binary, modified Heap data structure in Java where each parent node contains 2^x children nodes.

- **Implementation:** Authored `PowerOfTwoMaxHeap.java` with dynamic child node index calculations based on user parameter x .
- **Optimization:** Reduced tree depth for high-fanout operations, ensuring optimal $O(\log n)$ insertion and `popMax` routines.

Task 2: Software Architecture & Design Patterns

Objective: Architect a decoupled, reconfigurable backend data processing system adhering to enterprise software engineering standards.

- **Implementation:** Applied the **Strategy Design Pattern** and **SOLID Principles** in `DataProcessorSystem.java`.
- **Features:** Built dynamic execution modes (`DUMP` , `PASSTHROUGH` , `VALIDATE`) and pluggable connectors for PostgreSQL, Redis, and Elasticsearch.

Task 3: Relational Database Design (3NF Schema)

Objective: Design a Third Normal Form (3NF) relational database schema for Walmart's retail logistics and Pet Department.

- **Implementation:** Created `schema.sql` modeling primary entities (`product` , `shipment` , `shipment_item`).
- **Outcome:** Enforced data integrity via Primary and Foreign Key constraints while removing insertion, update, and deletion anomalies.

Task 4: Data Munging & Python ETL Pipeline

Objective: Parse, transform, and merge disparate raw CSV data into a clean SQLite relational database.

- **Implementation:** Developed `Data_Munger.py` leveraging Python's native `sqlite3` and `csv` modules.
- **Resilience:** Integrated a custom `safe_int()` parsing function to handle missing/dirty fields without script termination, populating `shipping.db` seamlessly.

3. KEY LEARNINGS & OUTCOMES

- Mastered memory-efficient traversal and parent-child navigation in multi-child heap structures.
- Practiced object-oriented design principles to enable pluggable backend microservices.
- Applied standard database normalization techniques (3NF) to enterprise-level schema design.
- Constructed crash-resilient ETL scripts for messy, unstructured datasets in Python.